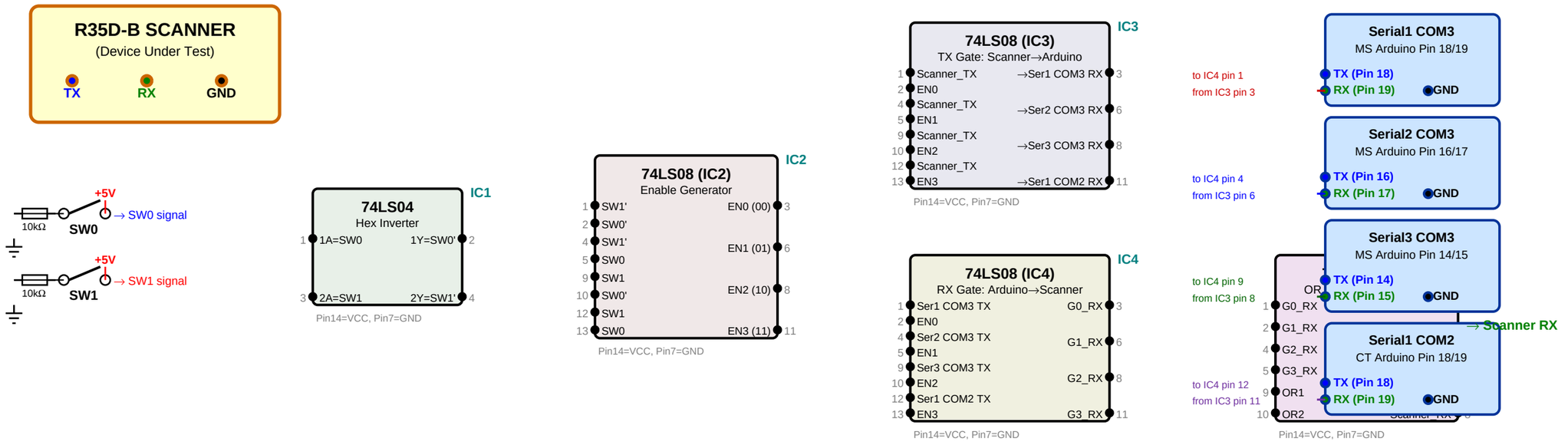
SERIAL PORT SELECTOR — WIRING DIAGRAM

4-to-1 MUX using 74LS08 (AND), 74LS32 (OR), 74LS04 (INV) | 2 Switches select 1 of 4 serial ports



LEGEND

SW0=0, SW1=0 → Serial1 COM3 (MS Arduino, Pin 18 TX / Pin 19 RX)

SW0=1, SW1=0 → Serial2 COM3 (MS Arduino, Pin 16 TX / Pin 17 RX)

SW0=0, SW1=1 → Serial3 COM3 (MS Arduino, Pin 14 TX / Pin 15 RX)

SW0=1, SW1=1 → Serial1 COM2 (CT Arduino, Pin 18 TX / Pin 19 RX)

SELECTION TABLE

SW1	SW0	EN Active	Port	Connection
0	0	EN0	Serial1 COM3	MS Arduino Pin 18/19
0	1	EN1	Serial2 COM3	MS Arduino Pin 16/17
1	0	EN2	Serial3 COM3	MS Arduino Pin 14/15
1	1	EN3	Serial1 COM2	CT Arduino Pin 18/19

BILL OF MATERIALS

1x 74LS04 (Hex Inverter) — IC1
3x 74LS08 (Quad 2-input AND) — IC2, IC3, IC4
1x 74LS32 (Quad 2-input OR) — IC5
2x Toggle Switch (SPST)
2x 10kΩ Resistor (pull-down)
5x 0.1µF Ceramic Capacitor (bypass)
1x R35D-B QR Scanner

IMPORTANT NOTES:

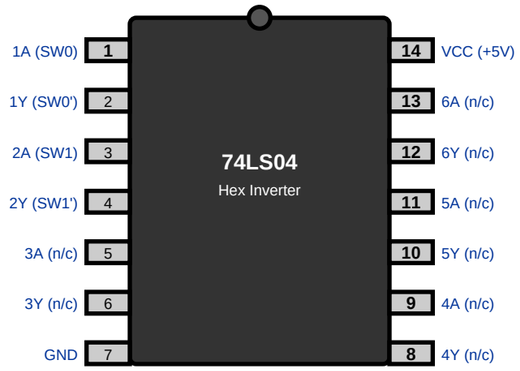
1. All ICs: VCC (Pin 14) = +5V, GND (Pin 7) = Ground. Place 0.1µF bypass cap between VCC and GND on each IC.
2. Switches: SPST toggle. One side to +5V, other side to signal line. 10kΩ pull-down to GND on signal side.
3. All GNDs must be connected together: Scanner GND + Arduino GND + IC GND = common ground.
4. 74LS series is 5V TTL. R35D-B scanner must be 5V TTL compatible. For 3.3V logic, add level shifters.

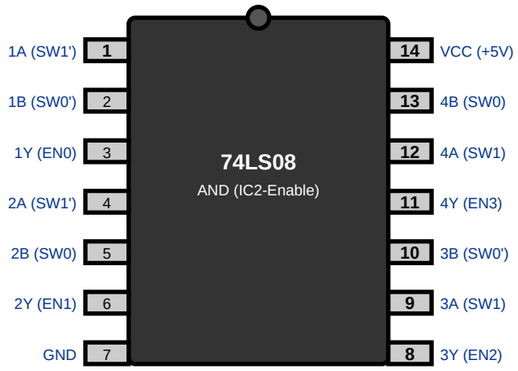
DETAILED PIN-TO-PIN WIRING LIST

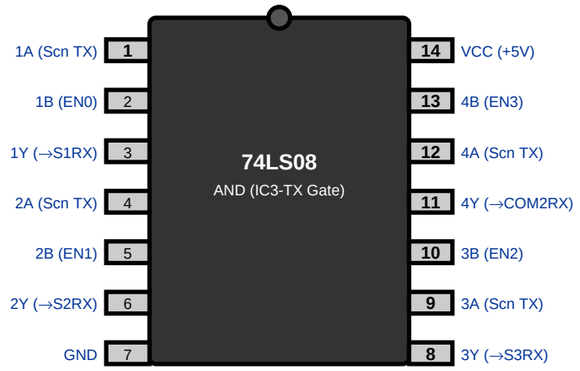
Every wire connection listed: FROM → TO

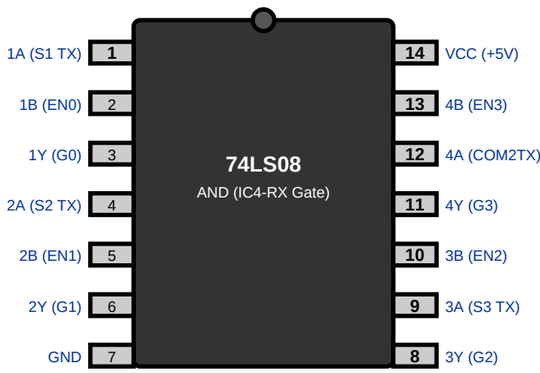
#	FROM		TO	PURPOSE
1	SW0 (switch output)	→	IC1 74LS04 Pin 1 (1A)	SW0 signal to inverter input
2	SW1 (switch output)	→	IC1 74LS04 Pin 3 (2A)	SW1 signal to inverter input
3	IC1 74LS04 Pin 2 (1Y=SW0')	→	IC2 74LS08 Pin 2, Pin 10	SW0 inverted to EN0 and EN2 generator
4	IC1 74LS04 Pin 4 (2Y=SW1')	→	IC2 74LS08 Pin 1, Pin 4	SW1 inverted to EN0 and EN1 generator
5	SW0 (direct)	→	IC2 74LS08 Pin 5, Pin 13	SW0 direct to EN1 and EN3 generator
6	SW1 (direct)	→	IC2 74LS08 Pin 9, Pin 12	SW1 direct to EN2 and EN3 generator
7	IC2 Pin 3 (EN0)	→	IC3 Pin 2, IC4 Pin 2	EN0 enables Serial1 COM3 gates
8	IC2 Pin 6 (EN1)	→	IC3 Pin 5, IC4 Pin 5	EN1 enables Serial2 COM3 gates
9	IC2 Pin 8 (EN2)	→	IC3 Pin 10, IC4 Pin 10	EN2 enables Serial3 COM3 gates
10	IC2 Pin 11 (EN3)	→	IC3 Pin 13, IC4 Pin 13	EN3 enables Serial1 COM2 gates
11	R35D-B Scanner TX	→	IC3 Pin 1, 4, 9, 12	Scanner TX to all 4 AND gate inputs
12	IC3 Pin 3 (gate out)	→	Serial1 COM3 RX (MS Arduino Pin 19)	Scanner data to Serial1 COM3
13	IC3 Pin 6 (gate out)	→	Serial2 COM3 RX (MS Arduino Pin 17)	Scanner data to Serial2 COM3
14	IC3 Pin 8 (gate out)	→	Serial3 COM3 RX (MS Arduino Pin 15)	Scanner data to Serial3 COM3
15	IC3 Pin 11 (gate out)	→	Serial1 COM2 RX (CT Arduino Pin 19)	Scanner data to Serial1 COM2
16	Serial1 COM3 TX (MS Arduino Pin 18)	→	IC4 Pin 1	Arduino trigger cmd to gate
17	Serial2 COM3 TX (MS Arduino Pin 16)	→	IC4 Pin 4	Arduino trigger cmd to gate
18	Serial3 COM3 TX (MS Arduino Pin 14)	→	IC4 Pin 9	Arduino trigger cmd to gate
19	Serial1 COM2 TX (CT Arduino Pin 18)	→	IC4 Pin 12	Arduino trigger cmd to gate
20	IC4 Pin 3 (G0_RX)	→	IC5 74LS32 Pin 1	OR tree input 1
21	IC4 Pin 6 (G1_RX)	→	IC5 74LS32 Pin 2	OR tree input 2
22	IC4 Pin 8 (G2_RX)	→	IC5 74LS32 Pin 4	OR tree input 3
23	IC4 Pin 11 (G3_RX)	→	IC5 74LS32 Pin 5	OR tree input 4
24	IC5 Pin 3 (OR1)	→	IC5 Pin 9	First OR output to final OR input A
25	IC5 Pin 6 (OR2)	→	IC5 Pin 10	Second OR output to final OR input B
26	IC5 Pin 8 (final OR out)	→	R35D-B Scanner RX	Combined Arduino TX to scanner
27	+5V VCC	→	IC1-IC5 Pin 14	Power all ICs
28	GND	→	IC1-IC5 Pin 7	Ground all ICs
29	GND	→	R35D-B Scanner GND	Common ground with scanner
30	GND	→	Arduino GND (COM3 + COM2)	Common ground with both Arduinos

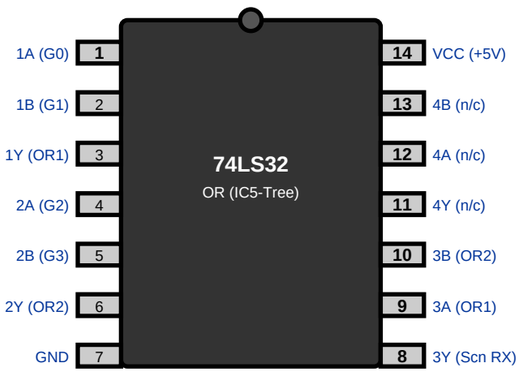
IC PINOUT REFERENCE











R35D-B SCANNER

QR Code Reader

- **TX (Yellow wire)**
  Scanner transmits scanned data
- **RX (Green wire)**
  Scanner receives trigger/stop cmds
- **GND (Black wire)**
  Common ground
- **VCC (Red wire)**
  Power +5V (from relay)